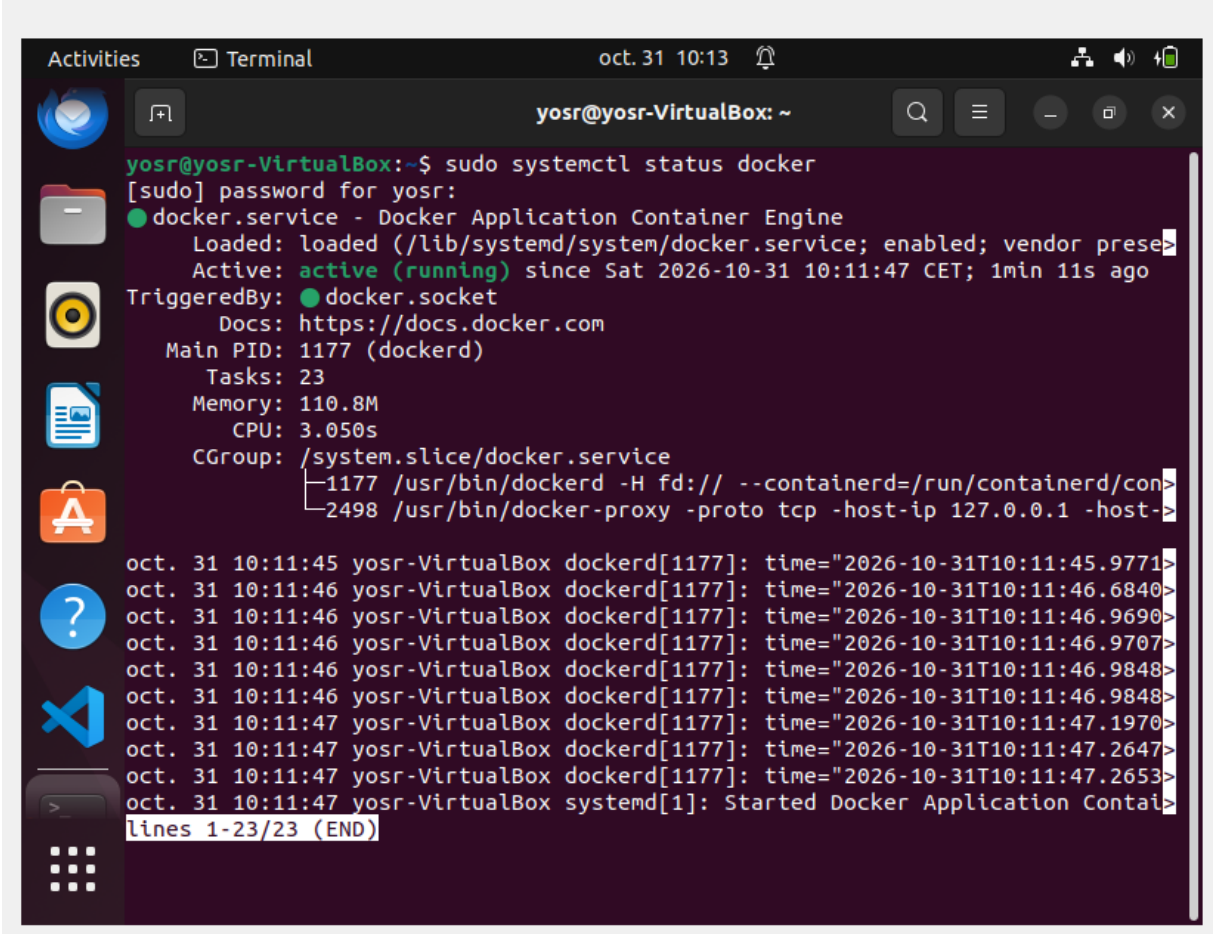
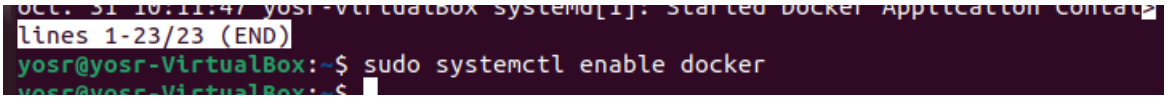
Compte rendu LAB03

1. Installation de Docker sous Ubuntu

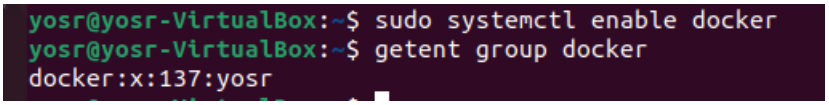
Pour vérifier l'état du service on tape :



Pour activer le service docker au démarrage système on tape :



Puisque la machine appartient déjà à un groupe, on vérifie avec cette commande :



2. Utilisation de la commande "docker"

Pour afficher toutes les sous-commandes disponibles, on tape :

```
yosr@yosr-VirtualBox:~$ docker
Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Authenticate to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
```

Pour afficher les options disponibles pour une commande spécifique, on tape :

```
For more help on how to use Docker, head to https://docs.docker.com/go/guides/
yosr@yosr-VirtualBox:~$ docker  run --help
Usage:  docker run [OPTIONS] IMAGE [COMMAND] [ARG...]

Create and run a new container from an image

Aliases:
  docker container run, docker run

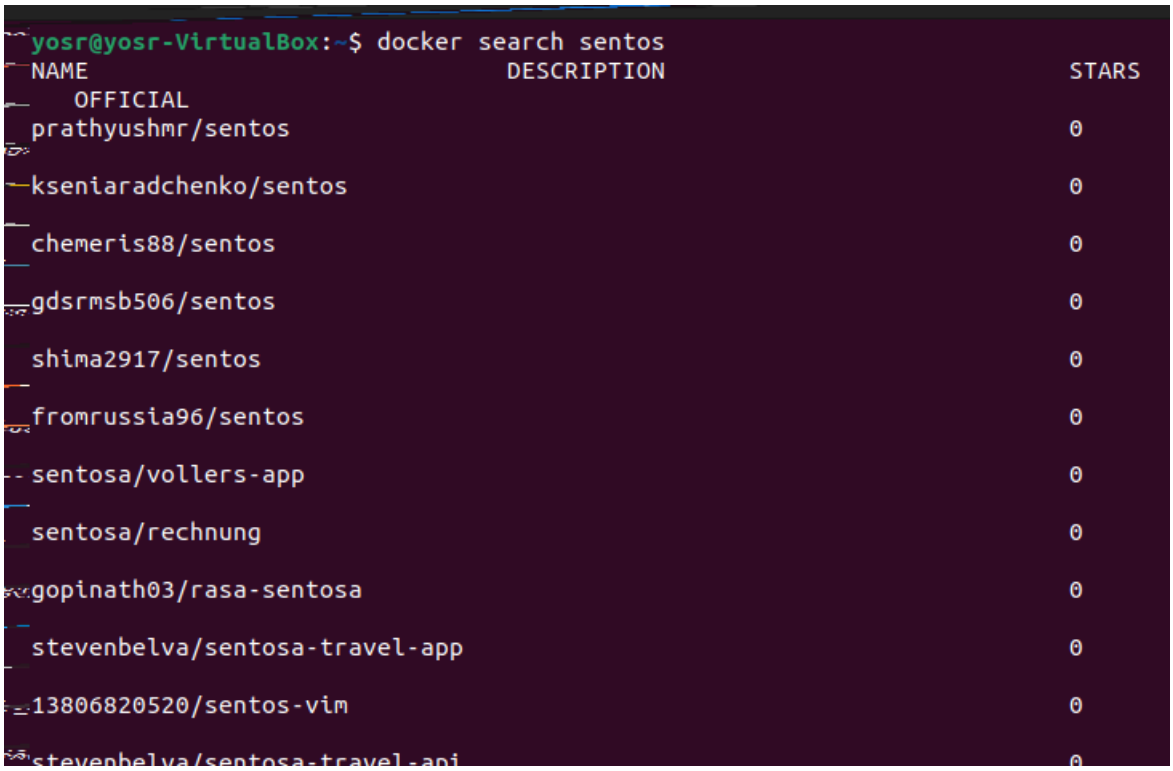
Options:
      --add-host list                  Add a custom host-to-IP mapping
                                       (host:ip)
      --annotation map                 Add an annotation to the
                                       container (passed through to
                                       the OCI runtime) (default map[])
  -a, --attach list                    Attach to STDIN, STDOUT or STDERR
      --blkio-weight uint16            Block IO (relative weight),
                                       between 10 and 1000, or 0 to
                                       disable (default 0)
      --blkio-weight-device list       Block IO weight (relative
                                       device weight) (default [])
      --cap-add list                   Add Linux capabilities
      --cap-drop list                  Drop Linux capabilities
```

Gestion des images

Pour voir la liste des images disponibles en local, on tape la commande :

```
                                    container
yosr@yosr-VirtualBox:~$ docker images
REPOSITORY              TAG       IMAGE ID       CREATED         SIZE
caddy                   latest    1b7d0a82297a   22 months ago   48.5MB
syslifters/sysreptor    2024.96   acee5d4b259e   23 months ago   2.66GB
postgres                14        365d502a72f3   23 months ago   422MB
bitnami/redis           7.2       3d7539dd4cb6   2 years ago     144MB
onosproject/onos        latest    2a4338117eab   2 years ago     876MB
onosproject/onos        2.7.0     92ab89ff82d2   4 years ago     836MB
```

Pour chercher une image dans le registre Docker, on tape :



Pour chercher toutes les images contenant le mot-clé centos et disposant d'au moins cinq étoiles en notation on utilise la commande suivante :

```
yosr@yosr-VirtualBox:~$ docker search --filter=stars=5 centos
NAME                            DESCRIPTION
STARS     OFFICIAL
centos                          DEPRECATED; The official build of CentOS.
7780      [OK]
centos/postgresql-10-centos7    PostgreSQL is an advanced Object-Relational …
21
centos/mongodb-36-centos7       MongoDB NoSQL database server
12
centos/nginx-112-centos7        Platform for running nginx 1.12 or building …
16
centos/mysql-57-centos7         MySQL 5.7 SQL database server
94
centos/httpd-24-centos7         Platform for running Apache httpd 2.4 or bui…
46
centos/systemd                  systemd enabled base container.
114
centos/mariadb-101-centos7      MariaDB 10.1 SQL database server
13
centos/postgresql-96-centos7    PostgreSQL is an advanced Object-Relational …
45
centos/redis-32-centos7         Redis in-memory data structure store, used a…
6
centos/mysql-56-centos7         MySQL 5.6 SQL database server
23
centos/nginx-18-centos7         Platform for running nginx 1.8 or building n…
14
centos/php-56-centos7           Platform for building and running PHP 5.6 ap…
```

Pour télecharger une image, par exemple centos on utilise :

```
yosr@yosr-VirtualBox:~$ docker pull centos
Using default tag: latest
Error response from daemon: manifest for centos:latest not found: manifest unkn
own: manifest unknown
```

La commande : docker pull centos équivaut à : docker pull centos:latest

Docker arrive à communiquer avec Docker Hub, mais il ne trouve pas l'image centos avec le tag latest. Or, l'image officielle CentOS ne propose plus de tag latest sur Docker Hub ce qui explique l'erreur affiché.

Pour résoudre ce problème on utilise cette solution pour télécharger une version précise :

```
yosr@yosr-VirtualBox:~$ docker pull quay.io/centos/centos:stream10
stream10: Pulling from centos/centos
9df62105b4c4: Downloading  41.38MB/67.17MB
bd9ddc54bea9: Download complete
```

Pour supprimer une image du dépôt local, sans conteneur démarré, on utilise la commande "rmi"

```
yosr@yosr-VirtualBox:~$ docker rmi  quay.io/centos/centos:stream10
Untagged: quay.io/centos/centos:stream10
Untagged: quay.io/centos/centos@sha256:99b8fe27dd4958b5dc2630ce35f5f6a7ff7bf9b2
b1a72078102fde50d3277550
Deleted: sha256:80414f28c1bee6bebd39bfcb9b19e7a57186f660c8baf9f112a004443ff338d
3
Deleted: sha256:ded8b2ebd083eebdba4325120e39758fb8340eaf7bd6ab69b0f45e2fff09111
3
Deleted: sha256:4c6c1f615688e5241ea4c1e508140da791ce7ea6216de985ef300972631 86a6
```

Gestion des conteneurs

- Pour créer un conteneur à partir d'une image :

```
quay.io/centos/centos:stream10
yosr@yosr-VirtualBox:~$ docker create --name MyFirstContainer centos:centos7
Unable to find image 'centos:centos7' locally
centos7: Pulling from library/centos
2d473b07cdd5: Pull complete

Digest: sha256:be65f488b7764ad3638f236b7b515b3678369a5124c47b8d32916d6487418ea4
Status: Downloaded newer image for centos:centos7
8f770e57374195194051dc36f2c14ac6ab2a0012267d49a0d99669c732be1e14
```

pour lister les conteneurs actifs :

```
yosr@yosr-VirtualBox:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
       STATUS                        PORTS                      NAMES
9ccd7128a06e   syslifters/sysreptor:2024.96   "/bin/bash /app/api/…"   21 month
s ago   Up About an hour (healthy)   127.0.0.1:8000->8000/tcp   sysreptor-app
d0638193962a   postgres:14                    "docker-entrypoint.s…"   21 month
s ago   Up About an hour (healthy)   5432/tcp                   sysreptor-db
bcbbe086d5de   caddy:latest                   "caddy run --config …"   21 month
s ago   Up About an hour                                        sysreptor-caddy
7182943615a7   bitnami/redis:7.2              "/opt/bitnami/script…"   21 month
s ago   Up About an hour             6379/tcp                   sysreptor-redis
```

Pour lister tous les conteneurs :

```
yosr@yosr-VirtualBox:~$ docker ps -a
CONTAINER ID   IMAGE                          COMMAND                  CREATED
             STATUS                      PORTS

                                        NAMES
8f770e573741   centos:centos7                 "/bin/bash"              About a
minute ago   Created

                                        MyFirstContainer
6053580109e7   onosproject/onos               "./bin/onos-service …"   12 month
s ago        Exited (255) 11 months ago   0.0.0.0:6653->6653/tcp, [::]:6653->66
53/tcp, 0.0.0.0:8101->8101/tcp, [::]:8101->8101/tcp, 6640/tcp, 9876/tcp, 0.0.0.
0:8181->8181/tcp, [::]:8181->8181/tcp   onos
a740eefc6219   onosproject/onos:2.7.0         "./bin/onos-service …"   12 month
s ago        Exited (255) 12 months ago   6640/tcp, 9876/tcp, 0.0.0.0:6655->665
3/tcp, [::]:6655->6653/tcp, 0.0.0.0:8103->8101/tcp, [::]:8103->8101/tcp, 0.0.0.
0:8183->8181/tcp, [::]:8183->8181/tcp   onos33
247a7183bf13   onosproject/onos:2.7.0         "./bin/onos-service …"   12 month
s ago        Created

                                        onos3
9244c7ed2f6b   onosproject/onos:2.7.0         "./bin/onos-service …"   12 month
s ago        Exited (255) 12 months ago   6640/tcp, 9876/tcp, 0.0.0.0:6654->665
3/tcp, [::]:6654->6653/tcp, 0.0.0.0:8102->8101/tcp, [::]:8102->8101/tcp, 0.0.0.
0:8182->8181/tcp, [::]:8182->8181/tcp   onos2
```

Pour démarrer un conteneur :

```
                                        sysreptor-redis
yosr@yosr-VirtualBox:~$ docker start 8f770e573741
8f770e573741
```

On va démarrer deux conteneurs :

```
yosr@yosr-VirtualBox:~$ docker run ubuntu uname -a
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
edafa32f5d9e: Pull complete
f3f25b84a7d7: Pull complete
Digest: sha256:6a06111c3b0a8600feadb270aec771dc239bf407b25625f93f44ab00e11f4d3c
Status: Downloaded newer image for ubuntu:latest
Linux c568584981a3 6.8.0-86-generic #87~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Mon
Sep 29 09:48:07 UTC 2 x86_64 GNU/Linux
yosr@yosr-VirtualBox:~$ docker run ubuntu uname -a
Linux 34e995284e59 6.8.0-86-generic #87~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Mon
Sep 29 09:48:07 UTC 2 x86_64 GNU/Linux
```

On vérifie la version du noyau :

```
yosr@yosr-VirtualBox:~$ uname -a
Linux yosr-VirtualBox 6.8.0-86-generic #87~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC M
on Sep 29 09:48:07 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
```

3. Image Docker, hello-world

Pour utiliser l'image "hello-word", on va l'exécuter sur un conteneur en utilisant la commande suivante :

```
yosr@yosr-VirtualBox:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:c567b231f65614688711a75f31518a064258a51d442117a3e771a3a8fdba78ef
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```